# ARISE

## Patient Flow Model

### A UX Design Proposal

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# Executive Summary

**Stop latency. Stop the manual chaos. Arise is here.**

Arise will improve patient flow in small medical clinics found throughout the country and rural communities. With Arise, you don't just track patients; you optimize the heartbeat of the clinic. By replacing fragmented communication with a high-octane, real-time Head's Up Display (HUD), Arise eliminates room latency and cognitive overload. This is the future of rural healthcare—confident, streamlined, and unmistakably powerful. Arise preserves the sacred doctor-patient relationship by taking the "tech stress" out of the equation by providing the following objectives:

## Objectives

- Reduce average room turnover time by 30%
- Provide instant status updates and push notifications.
- Eliminate manual whiteboard and sticky-note coordination
- Enable providers to access a 5-second patient summary before room entry
- Provide admin with daily flow export and weekly trend reports digitally
- Achieve color-blind accessibility

# Introduction

## Background

Small-town clinics are the backbone of community health, yet they are often held back by "whiteboard lag." When exam rooms sit empty while patients wait in the lobby, the clinic loses money and patients lose trust. Arise turns that idle time into active care.

## Problem Statement

Front Desk Staff, Clinical Staff, and Providers are currently operating in silos.

- **Blind Spots:** Clinical staff have to manually check the lobby.
- **Chart Diving:** Providers waste 15+ seconds per patient just finding the "why."
- **Data Fear:** Staff are terrified of "breaking the data" with one wrong click.
- **Ghost Operations:** Administrators have zero visibility into where the bottlenecks live.

# Objectives

**Arise isn't just a tool; it's a clinical navigation system.**

## Specific Goals

- **Front Desk Staff:** Experience lightning fast, one-click check-ins with an integrated insurance sidebar.
- **Clinical Staff:** Execute vitals with mobile-first precision. Haptic feedback confirms your work so you can keep your eyes on the patient.
- **Providers:** Walk into every room prepared. The Arise 5-Second Snapshot gives you the "need-to-knows" instantly.
- **Float Staff:** Hit the ground running. Arise's room map and quick add-on feature mean you don't need a week of training to be effective.

## Measurable Outcomes

- **30% Slash in Latency:** Watch room turnover move from "stagnant" to "streaming."
- **10-Second Handoffs:** 100% confidence in "Ready for Doctor" signals via haptic and visual alerts.
- **Zero-Panic Recovery:** 90% of errors reversed in under 5 seconds with our Arise Undo safety net.
- **Instant Reporting:** Go from 45 minutes of manual tallying to a 5-minute digital export.

# Design Requirements

## Functional Requirements

- **Bird's Eye View –** Real-time, color-coded room status (Clean, In Progress, Ready for MD, Wait Alert, Cleaning, Procedure). Every status has a unique shape glyph (◆, ▲, ■, ⚠, ●, ✦). No labels needed – instant recognition.

- **One-Click Check-In –** From the "Scheduled Arrivals" list, a single click pulls insurance and copay into a sidebar and moves the patient to "Ready to Room". No menus. No delays.

- **Undo Notification –** After any state change (check-in, room assignment, status update), a 5-second pop-up appears with a clickable "Undo". Mistakes vanish. Confidence stays.

- **Push Notifications –** The moment front desk staff assigns a patient to a room; the clinical staff's tablet gets a silent alert (vibration + banner). No shouting. No walking.

- **5-Second Snapshot –** Patient name, age, chief complaint, abnormal vitals (highlighted), and current wait time – all on one screen. Providers walk in informed.

- **"Ready for Doctor" Handoff –** One tap changes room status to "Ready" and alerts the provider on their tablet or wall display. No uncertainty. No wasted steps.

- **Next 3 Queue –** Provider sees the next three patients waiting in the lobby. Mental preparation becomes effortless.

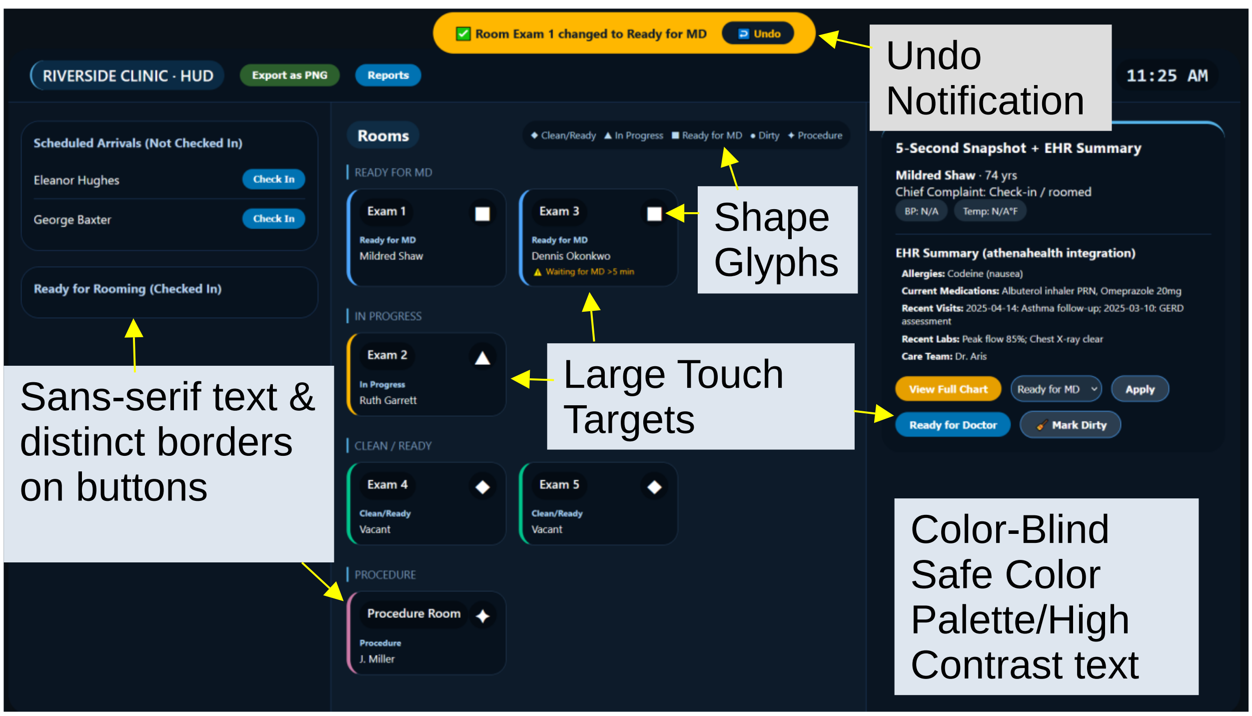
- **Daily Flow Export –** One click downloads a report of every room status change, wait time, and handoff timestamp. Reporting goes from 45 minutes to 5 seconds.

- **Walk-In Override –** Float staff adds an unscheduled patient with a temporary card, manual timer, and amber wait alert. No credentials are required. No chaos.

## Technical Requirements

- **Electronic Health Record (EHR) Agnostic –** Works with any system (athenahealth, eClinicalWorks, or a custom middleware layer). No vendor lock-in.
- **Multidevice –** Desktop for front desk, tablet for clinical, tablet/wall mount for providers, laptop for admin. One HUD, every role.
- **Offline First –** Network drops? The HUD caches the last known status and auto-syncs when connection returns. No downtime.
- **Role-Based Access –** Admin, clinical, provider, manager – each sees only what they need. And no auto-lock during active patient exams (non-negotiable for providers).
- **Clean Export Format –** Report with timestamp, room, previous status, new status, user role, and anonymized patient ID. Ready for your analytics stack.

## Aesthetic Requirements



# Methodology

## Research

**Arise was not built in a vacuum. Arise was envisioned in the hallways of real clinics and built from the experience of real professionals.**

- **Contextual inquiry:** We watched the rush. We felt the friction. (see Appendix C for *Stakeholder Questions* and *Requirements Gathering Q&A*)
- **Journey mapping workshops:** Co-created journey maps for each persona (see Appendix A for *Personas*).
- **Heuristic evaluation:** We mapped the system against Nielsen's principles to ensure "Visibility of System Status" is never a question.
- **Accessibility audit:** Used Stark and Color Contrast Analyser to define Wong palette and shape glyph requirements.

Key insights from personas (see Appendix A for detailed *Personas*):

- Front Desk Staff needs a **"quiet assistant"** — technology that stays in the background, auto-opens insurance, and has an unmistakable Undo.
- Clinical Staff require **fat-finger-friendly vitals entry** and haptic feedback to confirm saves in a noisy clinic.
- Providers demand a **5-second summary** and hate screen-staring; they want the Arise Model to be read-only during consultation.
- Float staff needs **self-explaining spatial layout** and phonetic search to avoid mismatching patients.
- Admin needs **exportable flow data** and trend lines without manual spreadsheet rebuilding.

## Design Process (see Appendix D for attached charts for Design Process)

1. **Low-fidelity wireframes:** Sketched Bird's Eye View, check-in sidebar, and snapshot panel. Tested with Front Desk/Admin for layout mimicry.
2. **Interactive prototype (mid-fi):** Built HTML/CSS/JS prototype with Wong palette, undo toast, and room grouping by status. Conducted cognitive walkthroughs with each persona.

3. **High-contrast accessibility pass:** Added shape glyphs, dark background, and increased touch target sizes to 44x44px.
4. **Usability testing (planned):** 5 participants representing each persona will complete core tasks (check-in, rooming, handoff, undo, export).

## Tools and Technologies

- **Prototyping:** HTML5, CSS3, JavaScript (html2canvas for export).
- **Design & collaboration:** Figma for high-fidelity mockups, Miro for journey mapping.
- **Accessibility testing:** Chrome DevTools (color blindness simulator), WCAG Contrast Checker.
- **Version control:** GitHub for code and document history.

# Conclusion

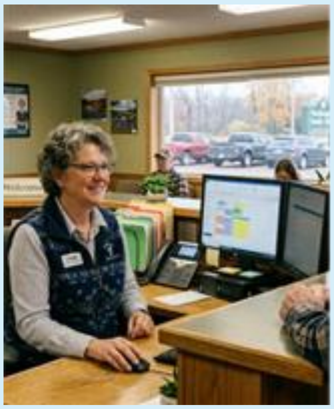
Arise is the missing link between modern efficiency and rural community trust. **It's time to move past the sticky notes.**

- Review the personas.
- Validate the EHR integration.
- Walk through the HUD prototype.

**Let's Arise together.**

# Appendices

## Appendix A: Personas & Journey Maps



### Linda

**Office Manager / Front Desk**

#### About

Age: 58

Rural Clinic Manager

Rural Clinic

Community-Oriented

Primary: Desktop

Status: 20-Yr Clinic Veteran

[Empathetic]

[Institutional]

[Meticulous]

[Warm] [Steadfast]

#### Quote

*"I don't want to learn a computer. I want to look up and see Arthur's face, not a loading screen."*

### Motivations

Linda's core drive is maintaining the human warmth of a clinic she has nurtured for 20 years. She is motivated by the elimination of clutter-induced stress—freeing herself from smudged whiteboards and lost sticky notes so she can focus on what matters: the patients.

### Goals

- Ensure elderly patients feel heard and valued during every interaction.
- Reduce the 'chaos' of the morning rush through a calm, organized check-in flow.
- Maintain a real-time 'Bird's Eye View' of all room statuses to coordinate rooming.
- Eliminate paper files without losing the personal touch of her current system.

### Pain Points

- Fears technology will make the clinic feel cold and corporate.
- Dreads accidentally clicking the wrong patient and 'breaking the data.'
- Frustrated by page-load delays when a patient is standing at the desk.
- Overwhelmed by cluttered UIs with too many irrelevant technical alerts.

### Requirements

- Large, high-contrast text and clearly clickable buttons.
- Prominent 'Undo' toast notification for accidental actions.
- Bird's Eye View dashboard that mirrors the physical clinic layout.
- Auto-integrated insurance/co-pay sidebar on patient selection.

### Personality Profile

Introvert ◀———————————▶ Extrovert

Thinking ◀———————————▶ Feeling

Judging ◀———————————▶ Prospecting

### Technical Skills

| | |
|---|---|
| Institutional Knowledge | ★★★★★ |
| Paper-Based Filing | ★★★★★ |
| Conflict De-Escalation | ★★★★★ |
| Digital Systems | ★★☆☆☆ |
| EMR / EHR Tools | ★★☆☆☆ |

### Heuristic Tie

#### Error Prevention

Linda's greatest fear is 'breaking the system' with an accidental click. The HUD must provide a clear Undo path (toast notification) and confirm destructive actions before they happen. Error prevention means she never has to panic-call IT.

### Scenario

#### The Walk-In Override

An unscheduled patient arrives. Linda must integrate them into the live flow without disrupting the scheduled queue or losing her Bird's Eye View

#### The Rush

Linda manages a high-volume check-in flow for recurring patients, many of them elderly.

## Journey 1 — The Walk-In Override

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Walk-In Arrives | An unscheduled patient arrives during peak check-in. Linda searches the expected list and finds no match. | Disrupted but composed | Create a temporary walk-in record without leaving the main screen. | Must open full EHR in another window – loses real-time clinic view. |
| Triage Decision | Linda verbally assesses urgency, decides the patient can wait, and seats them. | Calm authority | Visual distinction between walk-ins and scheduled patients. | Walk-ins and scheduled patients appear mixed together – hard to track. |
| Wait Management | After 25 minutes, Linda notices the wait and proactively updates the patient. | Empathetic | Automatic wait-time tracking and alerting for manually added walk-ins. | Alert system ignores manually added walk-ins – only works for scheduled patients. |
| Room Assignment | A room opens. Linda directs the walk-in there, overriding the next scheduled patient. | Decisive | Record priority overrides in the flow log for later review. | No log of the override – change invisible to management. |
| Scheduled Recovery | Linda calls the displaced scheduled patient to reschedule and apologize. | Responsible | Clear flag on displaced patient to ensure follow-up without paper notes. | No flag – she writes a sticky note, bypassing the digital system. |

## Journey 2 — Linda's Rush

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Patient Arrival | A familiar patient approaches. Linda retrieves their record. | Focused but cautious | Auto-display insurance and co-pay upon patient selection. | Slow page loads while patient waits at the desk. |
| Check-In & Validation | Linda confirms insurance and marks patient as arrived. | Confident | Clear, immediate confirmation that patient is now in the waiting pool. | Small touch targets – easy to tap wrong button in a rush. |
| Rooming the Patient | A room becomes available. Linda assigns the waiting patient to it. | Empowered | Silent, automatic notification to clinical team of room assignment. | Worries that digital handoff didn't reach the nurse. |
| The “Oops” Moment | Linda accidentally marks the wrong patient as arrived. | Anxious | Simple, fast way to reverse the last action without admin help. | No undo – locked into error or must call IT. |
| Post-Rush Review | Morning flow ends. Linda reviews the overall activity. | Relieved & Satisfied | Save flow history (check-ins, assignments, wait times) for later reporting. | No flow history saved – all reporting must be manual. |

# Elena

**Lead Medical Assistant/Clinical Staff**

## About

Age: 31

Lead Medical Assistant

Rural Clinic

Bilingual (EN / ES)

Primary: Tablet

Energetic & Mobile

[Efficient]

[Adaptable]

[Bilingual]

[Tech-Savvy]

[Detail-Oriented]

## Quote

*"If I have to tap more than twice to get to vitals, that's two seconds I wasn't watching the patient."*

## Motivations

Elena's motivation is speed and precision. She sees the digital tool as her 'navigation system'—something that feeds her information proactively so she can minimize room latency and keep the doctor moving without unnecessary back-and-forth.

## Goals

- Minimize 'Room Latency': the time a room sits empty or dirty between patients.
- Receive instant push notifications when a patient is checked in by Linda.
- Capture vitals quickly using a mobile-first, fat-finger-friendly interface.
- Signal the doctor with a single unmistakable 'Ready for Provider' button.

## Pain Points

- Delayed sync: walking to the lobby only to find no patient waiting.
- Login timeouts mid-workflow requiring re-authentication.
- Small input fields requiring pinch-zoom on a moving tablet.
- Uncertainty whether the doctor actually received the 'patient ready' alert.

## Requirements

- Mobile-first tablet interface with large tap-friendly touch targets.
- Guided Vitals Flow screen: BP → Temp → Weight in locked sequence.
- Haptic feedback on successful save (clinic noise makes sound unreliable).
- Shared 'Bird's Eye View' map with color-coded room statuses.

## Personality Profile

Introvert ◄——► Extrovert

Thinking ◄——► Feeling

Judging ◄——► Prospecting

## Technical Skills

| | |
|---|---|
| Rapid Vitals Collection | ★★★★★ |
| EMR / EHR Charting | ★★★★☆ |
| Triage & Room Prep | ★★★★★ |
| Tablet / Mobile UX | ★★★★☆ |
| Bilingual Communication | ★★★★★ |

## Heuristic Tie

### Flexibility and Efficiency of Use

Elena needs accelerators like a large numeric keypad for vitals, haptic feedback for saves, and push notifications that come to her – not the other way around. The HUD should let a power user like her skip steps and move fast without breaking the workflow.

## Scenario

### End-of-Day Room Turnover

Elena manages the end-of-day cleaning sweep, marking rooms clean in sequence and flagging supply and status issues before the clinic closes — all through the HUD on her tablet.

### The Back-Office Shuffle

Elena moves patients from lobby to exam rooms.

## Journey 1 — The End-of-Day Room Turnover

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Last Patient Done | Dr. Aris signals checkout on the final patient of the day. | Tired but focused | Rooms should auto-update to “needs cleaning” when provider closes a chart. | Some rooms stay in “ready” status because provider forgot to tap checkout. |
| Room-by-Room Sweep | Elena moves through each exam room, disinfects, and marks it clean. | Methodical | Show her exact progress – how many rooms remain to be cleaned. | Map shows colors but no count – she must mentally track remaining rooms. |
| Supply Check | While cleaning Room 3, she notices the glove supply is low. Logs it. | Proactive | A simple field per room to flag supply issues for the manager. | No supply field – she texts Marcus directly, which may go unread. |
| Procedure Room | The procedure room requires extended cleaning. | Focused | A distinct status for “extended clean” to signal it won’t be ready soon. | Only “needs cleaning” status – looks identical to a 2-minute wipe-down. |
| EOD Confirmation | All rooms show clean. Elena screenshots the board for her record. | Satisfied | Automated end-of-day confirmation that logs to management with a timestamp. | No automated log – screenshot is her only proof of cleaning. |

## Journey 2 — The Back Office Shuffle

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Active Monitoring | Elena carries her device while finishing a task; sees a “new patient arrived” alert. | Alert & ready | Instant sync with front-desk check-in actions. | Delayed sync – she walks to the lobby only to find it empty. |
| Patient Pickup | Elena greets the patient and pulls up their intake form. | Focused | One-tap access to vitals intake – no extra login screens. | Session times out – she has to re-authenticate mid-workflow. |
| Data Entry | Elena inputs blood pressure, weight, and temperature. | Efficient | Large touch targets for data entry while walking and talking. | Small fields require pinch-to-zoom on a moving tablet. |
| The Handoff | She taps “ready for provider” and exits the room. | Relieved | Provider’s device should alert them immediately. | Uncertainty whether the provider actually received the alert. |
| Room Prep | Elena marks the vacated room as “needs cleaning” on the map. | Organized | Front desk sees the dirty status and won’t send the next patient there. | Front desk sends a patient to a room that hasn’t been disinfected. |

## Dr. Aris

**Owner / Primary Physician**

### About

Age: 45

Specialist (Dermatology)

Rural Clinic

Detail-Oriented

Tablet / Watch

Focus-Driven

[Decisive]

[Empathetic]

[Precise]

[Efficient]

[Patient-Centered]

### Quote

*"Tell me what I need to know before I walk through that door. After that, get out of my way."*

## Motivations

Dr. Aris wants to maximize the quality of every patient encounter. His internal drive is reclaiming face-time—spending more minutes on diagnosis and human connection and fewer on 'software logistics.' He views the tool as a silent HUD, not an EHR.

## Goals

- Achieve a '5-second summary': name, chief complaint, vitals on one screen.
- Move seamlessly between rooms without asking staff where to go.
- See 20+ patients daily without cognitive burnout from fragmented data.
- Dispatch lab/referral orders without leaving the exam room.

## Pain Points

- Dashboard shows Room 2, but the actual patient is somewhere else.
- Having to dig through 'History' tabs to find today's chief complaint.
- System auto-locks every 2 minutes, requiring a password mid-exam.
- Being 'blind' to lobby volume until it becomes a bottleneck crisis.

## Requirements

- Heads-Up Display (HUD): Patient Name + Reason + Wait Time on one screen.
- High-contrast badges for abnormal vitals (e.g., red circle on high BP).
- 'Next 3' lobby queue so he can mentally prepare for upcoming complexity.
- Smart Handoff prompt: 'Is this room ready for cleaning?' on chart close.

## Personality Profile

Introvert ◄──────► Extrovert

•

Thinking ◄──────► Feeling

•

Assertive ◄──────► Turbulent

•

## Technical Skills

| | |
|---|---|
| Advanced Diagnostics | ★★★★★ |
| Surgical Procedures | ★★★★★ |
| Patient Empathy | ★★★★★ |
| EMR / EHR Usage | ★★★☆☆ |
| Mobile / Wearable Tech | ★★★☆☆ |

## Heuristic Tie

### Recognition Rather Than Recall

Dr. Aris should not have to remember or search for key information. The 5-second Snapshot must surface the patient's name, chief complaint, abnormal vitals, and wait time on a single screen – so he can walk into the room already informed.

## Scenario

### The Behind-Schedule Recovery

Dr. Aris falls 25 minutes behind and uses the HUD to strategically reorder his patient queue, signal Linda, and recover the clinic's pace before the lunch break.

### The Clinical Pivot

Dr. Aris moves between exam rooms seamlessly using a HUD snapshot—eliminating chart-diving and staff interruptions so he can sustain high-quality face-time with every patient throughout the day.

## Journey 1 — The Behind-Schedule Recovery

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Delay Recognition | Dr. Aris notices he is running 25 minutes behind schedule. | Pressured | A clear readout of exactly how many minutes behind he is. | Health bar shows only a percentage – he can't translate it to actual delay. |
| Queue Prioritization | He reviews the upcoming patients and sees a complex case followed by two routine follow-ups. | Strategic | Ability to reorder the queue based on visit complexity and duration. | Queue is fixed by check-in time – no flexibility for clinical urgency. |
| Expedited Visit | He moves to a routine follow-up and completes it quickly. | Efficient | The patient summary should flag "short visit" or "complex" to help him self-select. | All visits look identical in the summary – no complexity indicator. |
| Signal to Staff | He sends a delay alert to the front desk for lobby management. | Communicative | One-tap delay signal that front desk sees immediately. | No delay signal – he must step out of the exam room to speak to staff in person. |
| Recovery Achieved | By lunch, he has caught up and finishes the morning session on time. | Relieved | System confirms he has "caught up" with a subtle acknowledgment. | Health bar just goes up – no sense that the system noticed his effort. |

## Journey 2 — Dr. Aris's High-Stakes Day

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| The Transition | Dr. Aris finishes with a patient and checks which room to go to next. | Pressured but focused | Know exactly which room to enter without asking staff. | Dashboard says Room 2, but the actual patient is somewhere else. |
| The Pre-Entry Brief | He scans the patient summary (reason, vitals, wait time) before entering. | Prepared | All critical "need-to-knows" on a single screen. | Must dig through history tabs to find today's chief complaint. |
| The Consultation | He sets the device down and focuses entirely on the patient. | Engaged | Tool stays quiet – no interruptions or lockouts during the exam. | System auto-locks every 2 minutes, requiring a password mid-exam. |
| The Order/Signal | At visit end, he orders labs or a referral, if needed. Otherwise, he closes chart. | Decisive | Notification goes directly to clinical staff without leaving the room. | Must physically leave the room to find staff for a task. |
| The Shift | He moves to the next patient and sees he is 15 minutes behind. | Mindful | Lobby volume and delay status visible at a glance. | Blind to lobby volume until it becomes a crisis. |

# Carmen Vega

**Float / Per-Diem Front Desk**

## About

Age: 26

Float Staff

Multiple Clinics

Bilingual EN/ES

Primary: Desktop

Adaptable

[Resourceful]

[Bilingual]

[Observant]

[Quick Learner]

[People-Oriented]

## Quote

*"If I can't figure it out in the first 10 seconds, I've already lost the patient's trust."*

## Motivations

Carmen thrives on variety and takes pride in seamlessly stepping into unfamiliar environments. She is motivated by the challenge of 'hitting the ground running' — arriving at a new clinic and earning the trust of permanent staff within the first hour. The HUD is her lifeline in an unfamiliar setting.

## Goals

- Orient to an unfamiliar clinic's HUD layout within 10 minutes of arrival.
- Serve Spanish-speaking patients confidently without slowing check-in flow.
- Avoid escalating issues to the permanent staff unless absolutely necessary.
- Leave the clinic in better shape than she found it — no data errors.

## Pain Points

- Every clinic's HUD is set up slightly differently — room layouts never match.
- No onboarding handoff — she is often briefed by a sticky note on the monitor.
- Unfamiliar patient names make it hard to confidently match walk-ins to records.
- Fear of making irreversible check-in or rooming errors in the first 30 minutes.

## Requirements

- Spatial HUD that is self-explaining — room labels and colors need no legend.
- Prominent Undo / correction path for accidental actions.
- Patient name search that tolerates phonetic spelling variants.
- Bilingual UI or tooltip support for patient-facing confirmation messages.

## Personality Profile

Introvert ◄——► Extrovert

Thinking ◄——► Feeling

Judging ◄——► Prospecting

## Technical Skills

| | |
|---|---|
| Front Desk Operations | ★★★★☆ |
| EHR / Check-In Systems | ★★★☆☆ |
| Bilingual Communication | ★★★★★ |
| HUD Orientation Speed | ★★★☆☆ |
| Conflict De-Escalation | ★★★☆☆ |

## Heuristic Tie

**Recognition over recall + Error prevention**

Carmen cannot rely on memory of a clinic she's never worked in. The HUD must make every action self-evident and every mistake recoverable.

## Scenario

**The First-Day Float**

Carmen needs to orient to an unfamiliar clinic's HUD within 10 minutes, check in patients accurately without errors, leaving the permanent staff confident that she didn't cause any data problems.

**The Angry Walk-In**

She needs to handle an unscheduled, frustrated patient without disrupting the scheduled queue.

## Journey 1 — The First-Day Float

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| First Patient | A returning patient arrives. Carmen searches by name and finds multiple matches. | Cautious | Phonetic search or photo confirmation to avoid selecting the wrong record. | Two patients with similar names – she must ask for birth date publicly. |
| Check-In & Insurance | She selects the patient; insurance and co-pay details appear. | Relieved | Auto-pull works the same for float staff as for permanent staff. | Sidebar fails to load because float credentials have different permissions. |
| Rooming | A room becomes available. She assigns the patient to it. | Empowered | Confirmation that clinical staff received the room assignment. | No confirmation – she doesn't know if the nurse saw the update. |
| Undo Recovery | She accidentally checks in the wrong patient, then spots the undo option. | Startled then calm | Undo should revert cleanly without leaving duplicate entries. | Undo works but leaves a ghost "arrived" status – she can't tell if it fully cleared. |
| End of Coverage | The regular staff returns. Carmen reviews her activity log. | Satisfied | A readable session summary she can show the returning staff. | No session log – she must manually brief from memory. |

## Journey 2 — The Angry Walk-In

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Walk-In Arrives | A visibly frustrated patient demands to be seen without an appointment. | Tense | A "walk-in" queue separate from the scheduled list. | No visible walk-in slot – she must interrupt the physician by phone to ask. |
| Creating the Record | She attempts to create a temporary walk-in entry. | Uncertain | A simple "add walk-in" button that creates a placeholder without full EHR credentials. | Walk-in creation requires EHR credentials that float staff don't have. |
| Lobby Management | She seats the patient and gives an estimated 20-minute wait. | Managing pressure | Automatic wait-timer alert so she remembers to check back. | No manual timer – she relies on her phone alarm as a workaround. |
| Escalation Signal | She sees multiple patients have been waiting based on the HUD signals. Notifies clinical staff. | Stressed | An in-app message or flag to staff about the waiting walk-in. | No in-HUD messaging – she must leave the desk to find staff. |
| Resolution | The patient is eventually seen. Carmen marks them as checked out. | Relieved | Checkout removes them cleanly from the active board. | Checkout leaves the room in "needs cleaning" – she's unsure if staff knows. |

## Marcus Webb

**Practice Administrator**

### About

Age: 58

Operations Manager

Rural Clinic

Data-Driven

Primary: Laptop

Analytical

[Strategic]

[Methodical]

[Efficiency-Focused]

[Policy-Minded]

[Detail-Oriented]

### Quote

*"I don't need to watch the HUD in real time. I need it to tell me, quietly, where we broke down last week."*

### Motivations

Marcus does not touch the HUD during patient hours — he views it from a distance, as an operational instrument. His motivation is using aggregated flow data to identify bottlenecks, reduce wait times, and make staffing decisions that Dr. Aris and Linda never have to think about. He is the person who makes the system work before anyone arrives.

### Goals

- Use end-of-day HUD flow data to identify recurring bottlenecks by room or time slot.
- Generate weekly clinic health reports without pulling data manually from the EHR.
- Detect staffing gaps before they become patient experience failures.
- Present operationally clear metrics to Dr. Aris in under 5 minutes per week.

### Pain Points

- The HUD has no report export — he rebuilds the data manually in a spreadsheet.
- Flow history resets daily; there is no persistent record of week-over-week trends.
- He has no visibility into which room has the highest latency without watching in real time.
- Dr. Aris asks questions Marcus can't answer without digging through the EHR.

### Requirements

- Daily flow history export (CSV or PDF) showing per-room and per-provider throughput.
- Week-over-week trend line for clinic health score and average wait time.
- A 'manager view' on the HUD: read-only, no accidental actions, with drill-down on delays.
- Automated weekly summary email generated from HUD data — no manual entry.

### Personality Profile

Introvert
◀————————————————▶
Extrovert
•
Thinking
◀————————————————▶
Feeling
•
Judging
◀————————————————▶
Prospecting
•

### Technical Skills

| | |
|---|---|
| Operations Management | ★★★★★ |
| Data Analysis / Reporting | ★★★★☆ |
| EHR Administration | ★★★★☆ |
| HUD Real-Time Use | ★★☆☆☆ |
| Staffing / Scheduling | ★★★★★ |

### Heuristic Tie

**Visibility of system status + Flexibility & efficiency**

Marcus needs the HUD to surface operational truth passively — without requiring him to watch it in real time. The data must come to him, not the other way around.

### Scenario

**The End-of-Week Audit**

To extract actionable operational data from the HUD – room latency, handoff delays, and throughput trends – without manual spreadsheet rebuilding, in order to present a clear weekly report to Dr. Aris in under 5 minutes.

**The Bottleneck Intervention**

To detect a stuck room or process failure in real time (without watching the HUD constantly), intervene remotely, to log the incident for future prevention.

## Journey 1 — The End-of-Week Audit: translating HUD data into operational decisions

| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Weekly Setup | Marcus opens the tool on Friday afternoon to pull flow data. | Methodical | One-click export of week-over-week flow data. | No reports tab – he screenshots the board and manually times each room. |
| Identifying Delays | He notices Room 2 has the longest average wait times all week. | Analytical | Sortable per-room latency view (time-in-status per patient). | Data not available at room level – only total clinic wait time visible. |
| Staffing Analysis | He cross-references rooming times with provider entry times. | Focused | Handoff metric: time from “ready for provider” to “provider entered.” | Handoff time not tracked – he can see both events but not the gap. |
| Report Draft | He compiles findings into a weekly brief for leadership. | Productive | One-click formatted PDF or CSV export. | Rebuilds report from scratch in a spreadsheet – 45 minutes of manual work. |
| Action Items | He assigns follow-up tasks (adjust scheduling, add buffers). | Decisive | Tasks assignable directly in the tool as pinned notes or flags. | No task layer – he emails follow-ups separately. |

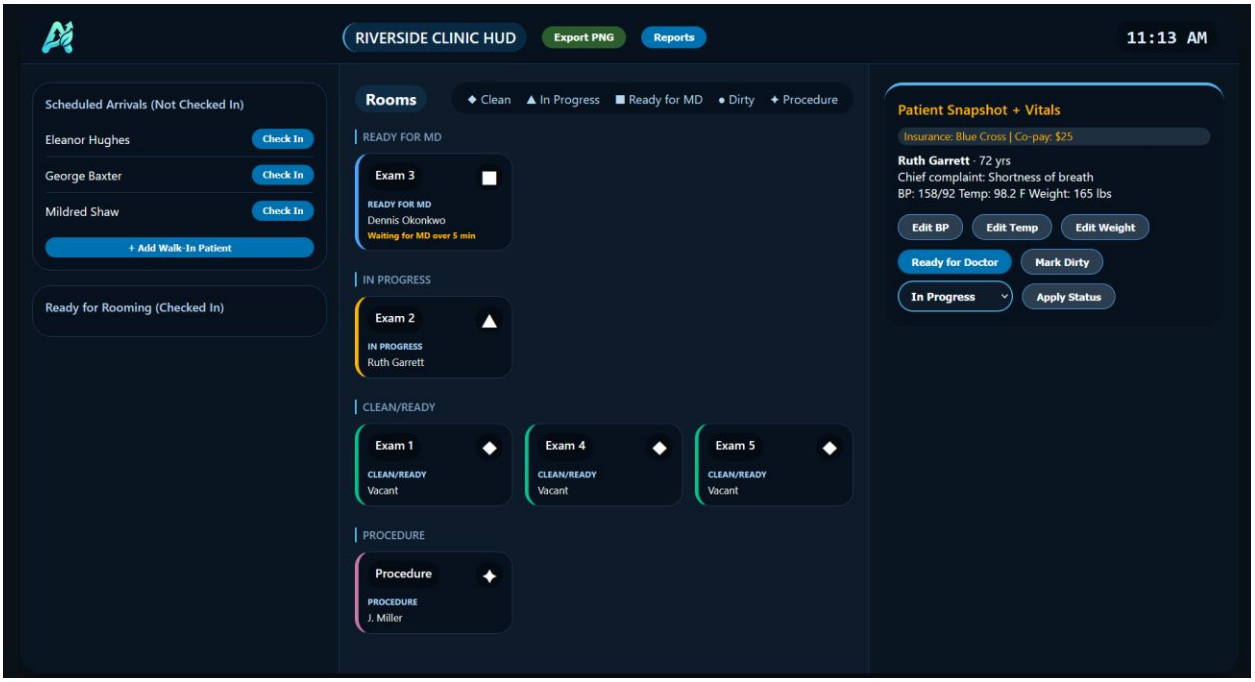
## Journey 2 — The Bottleneck Intervention: identifying and resolving a stuck room in real time

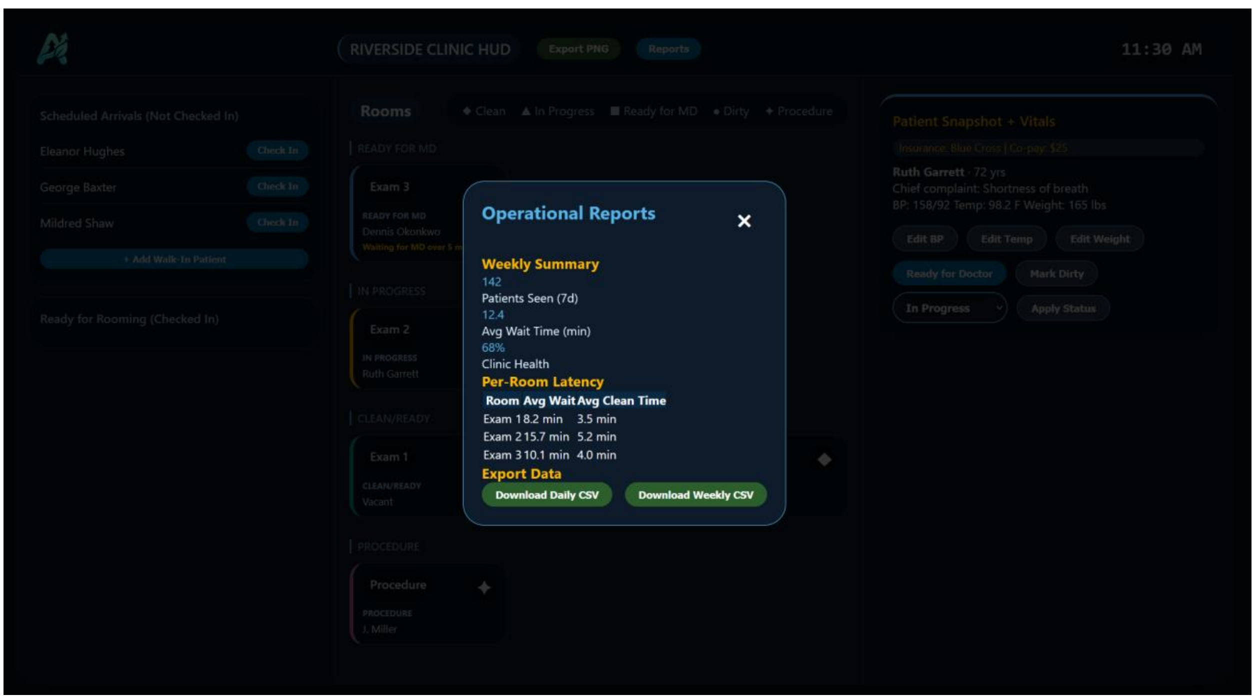
| Touchpoint | Action | Emotion | Expectation | Pain Point |
|---|---|---|---|---|
| Alert Detection | Marcus checks the dashboard and sees multiple waiting signals. | Concerned | Proactive threshold alert when multiple waiting signals have been triggered. | No proactive alert – he notices only because he happened to check. |
| Root Cause | He drills in and finds Room 4 has been “dirty” for 50 minutes. | Frustrated | A “stuck room” flag highlighting rooms exceeding normal duration. | No flag – room shows only static status; duration is not surfaced. |
| Intervention | He calls cleaning and notifies front desk to hold Room 4. | Taking control | An in-tool “hold” button to prevent new patients from being routed there. | No hold tool – front desk must remember not to assign Room 4. |
| Recovery Monitoring | He watches the waiting signals disappear over 45 minutes. | Relieved | Improvement is documented in the daily report. | No record of bottleneck. |
| Post-Incident Log | He documents the bottleneck cause, duration, and resolution. | Thorough | An incident log field that auto-timestamps events. | No incident log – he types a summary from memory into a shared doc. |

## Appendix B: High-Contrast HUD Prototype (HTML)

Interactive prototype demonstrating Bird's Eye View, undo toast, and snapshot panel.

[ARISE: Patient Flow Model]



## Appendix C: Stakeholder Questions & Requirements Gathering Q&A

**Stakeholder Questions:**

| Goal Area | Stakeholder Question & Resulting Constraint |
|---|---|
| **Front Desk: Cognitive Continuity** | **Question:** What elements must match your physical layout?<br>**Constraint:** HUD must mirror the physical clinic layout (Bird's Eye View). |
| **Front Desk: Integrated Verification** | **Question:** What info must appear automatically during check-in?<br>**Constraint:** Insurance and co-pay balance must auto-pull into a sidebar. |
| **Medical Assistant: Vitals Workflow** | **Question:** What steps are easiest to forget while multitasking?<br>**Constraint:** A guided flow is required: BP → Temp → Weight in a locked sequence. |
| **Medical Assistant: Provider Handoff** | **Question:** What makes the "Ready" signal unmistakable?<br>**Constraint:** Elena needs haptic feedback (vibration) to confirm a successful save. |
| **Doctor/Provider: Task Signaling** | **Question:** What tasks need to be streamlined at visit end? |

| Goal Area | Stakeholder Question & Resulting Constraint |
| --- | --- |
| | **Constraint:** Single-tap buttons for Lab or Referral orders from inside the room. |
| **Doctor/Provider: Time Management** | **Question:** What helps you understand when you're falling behind?<br><br>**Constraint:** A "Clinic Health" delay bar to visualize schedule deficits in real-time. |
| **Admin: Data Utility** | **Question:** What data do you need for weekly reports?<br>**Constraint:** Daily CSV export containing room timestamps, wait times, and throughput. |

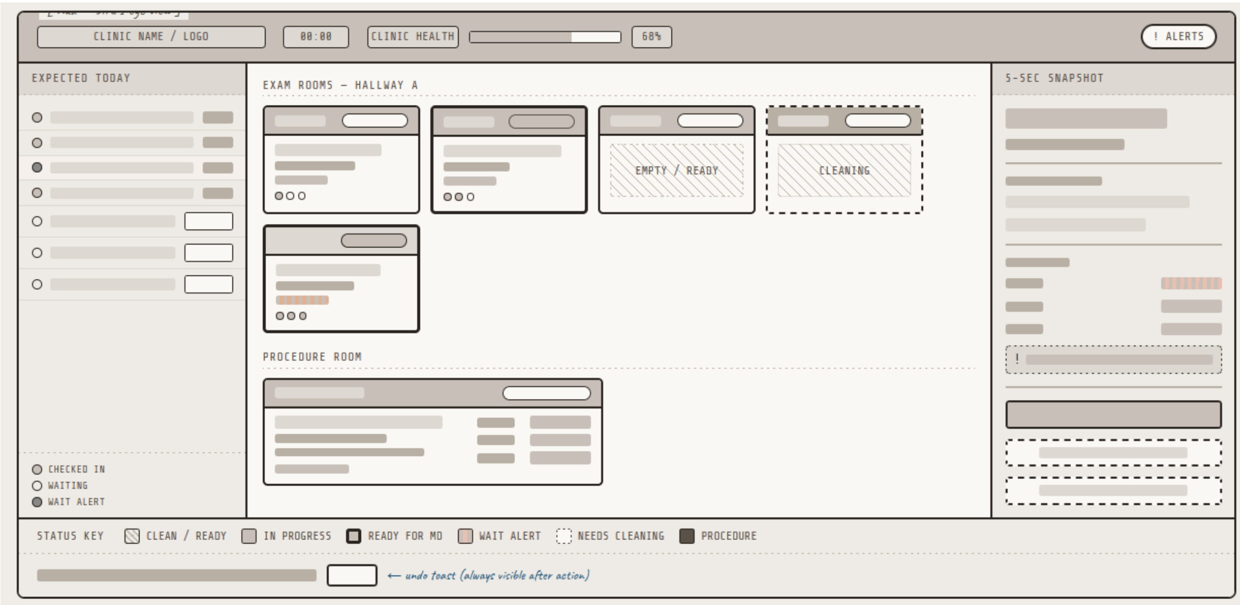
**Requirements Gathering Q&A:**

| Question / Topic | Answer / Constraint |
| --- | --- |
| **Current EHR System** | Must be EHR-agnostic using FHIR R4 or specific athenahealth/eClinicalWorks APIs. |

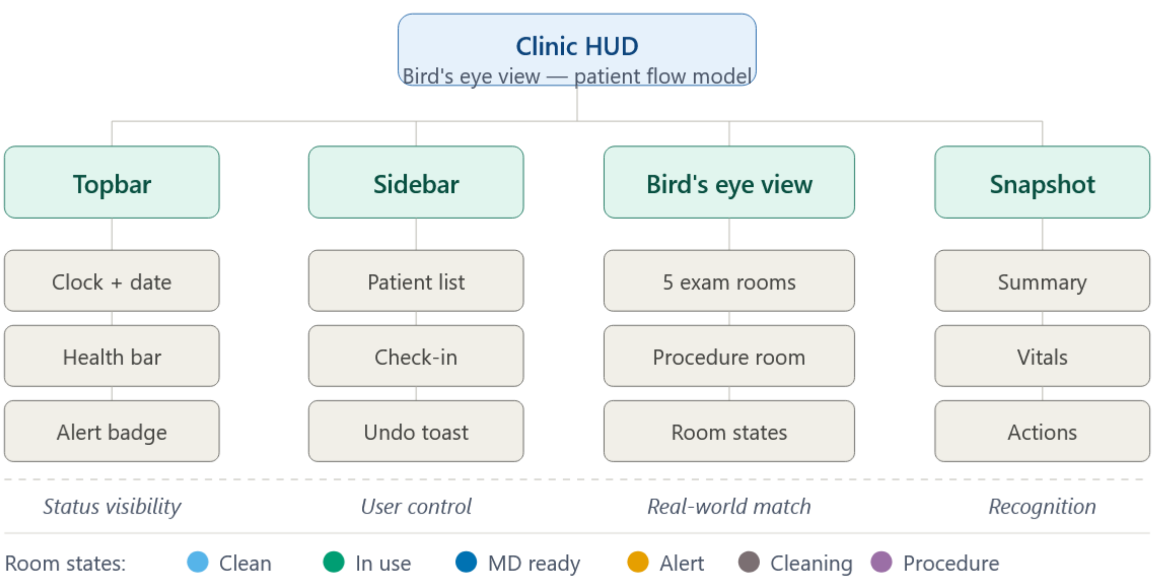
| Question / Topic | Answer / Constraint |
| --- | --- |
| **Integration Constraints** | Middleware layer required to translate data; must support offline fallback if network drops. |
| **Check-In Reliability** | Must be a one-click interaction from the "Expected Today" list to maintain eye contact. |
| **Error Recovery Trust** | Requires a 5-second Undo toast notification that appears after any state change. |
| **Mobile Adaptability** | UI must be "fat-finger-friendly" with minimum 44x44px touch targets; no stylus required. |
| **Provider Pre-Entry** | Requires a 5-second snapshot (Name, Complaint, Vitals, Wait Time) on a single screen. |

## Appendix D: WireFrames and Other Artifacts from Design Process

### WireFrame




### Heuristic SiteMap




*Patient flow model — clinic HUD site map*

## MoSCoW Chart & Priorities

| Feature / Requirement | Persona(s) | UX Rationale |
|---|---|---|
| **M — Must Have** *Core features without which the product fails* | | |
| **Bird's Eye View Dashboard** | Linda / Dr. Aris | Linda's core mental model mirrors her physical whiteboard. Dr. Aris uses it as his hallway HUD to know which room is ready. Without this, the entire coordination layer collapses. |
| **One-Click Patient Check-In** | Linda | Linda cannot navigate menus while making eye contact with Mrs. Higgins. A single-click check-in is the difference between a warm greeting and a cold transaction. |
| **Real-Time Room Status (Color Codes/Shape Glyphs)** | Linda / Elena / Dr. Aris | All three users depend on color coding and shape glyphs for room states. This is the shared language of the clinic's flow — removing it breaks every handoff. |
| **Integrated Insurance/Balance Sidebar** | Linda | Auto-displaying insurance and co-pay on patient selection lets Linda maintain eye contact. Forcing her to hunt for this data introduces delay and frustration at the front desk. |
| **Prominent 'Undo' Button (Toast)** | Linda | Linda's greatest fear is 'breaking the data.' A persistent, non-intrusive undo toast is her emotional safety net — without it, she may avoid the system entirely. |
| **Push Notifications (Patient Ready)** | Elena | Elena cannot return to the front desk between every task. Real-time push alerts are the engine of her back-office efficiency and prevent wasted trips. |
| **Guided Vitals Flow Screen** | Elena | A locked BP → Temp → Weight sequence prevents Elena from skipping steps while multitasking. It acts as a cognitive checklist inside a noisy, fast-paced environment. |
| **5-Second Snapshot (HUD View)** | Dr. Aris | Dr. Aris needs Name + Chief Complaint + Vitals before he enters a room. If this screen requires more than a glance, it defeats its own purpose. |
| **'Ready for Doctor' Button** | Elena / Dr. Aris | The single most critical handoff trigger in the clinic. Elena needs confidence her signal was sent; Dr. Aris needs certainty before he walks in. |
| **S — Should Have** *High-value features that significantly improve UX* | | |
| **Wait Time Alert** | Linda | When a patient’s wait signal (20+ min wait), Linda gets a soft visual cue to proactively manage lobby tension — fulfilling her goal of making patients feel heard. |
| **Drag-to-Room Assignment** | Linda | Dragging a patient name to a room icon reinforces the spatial metaphor of her whiteboard. It reduces the learning curve and makes rooming feel intuitive. |
| **Automatic Session Keep-Alive** | Dr. Aris / Elena | Auto-lock during a live exam forces Dr. Aris to re-authenticate mid-consultation. A session keep-alive tied to room occupancy prevents this disruption entirely. |
| **'Needs Cleaning' Room Status** | Elena / Linda | When Elena marks a room dirty, Linda must see it instantly to prevent routing a new patient into an unclean space — a patient safety issue masquerading as a UX feature. |
| **Lab / Referral Order Button** | Dr. Aris | Tapping 'Lab Needed' from inside the exam room eliminates the need to find Elena physically. This is a core time-recovery feature for a 20+ patient day. |

| | | |
|---|---|---|
| **Large-Format Numeric Keypad** | **Elena** | Standard keyboard input on a moving tablet causes mis-taps. A purpose-built number pad for vitals entry is table stakes for a mobile-first clinical interface. |
| **Smart Handoff Prompt** | **Dr. Aris** | When Dr. Aris closes a patient chart, a prompt — 'Is this room ready for cleaning?' — closes the loop with Elena automatically without requiring extra steps. |
| **C — Could Have** *Desirable enhancements if capacity allows* | | |
| **Voice-to-Text Vitals Entry** | **Elena** | Bilingual voice input would further speed Elena's workflow, particularly with gloved hands or when managing a difficult patient interaction simultaneously. |
| **Haptic Feedback on Save** | **Elena** | In a noisy clinic, a subtle tablet vibration confirms a successful save, letting Elena put the device down and move on without visual confirmation. |
| **Wall-Mounted Clinic Map Display** | **Dr. Aris** | A passive, always-on hallway monitor showing the Bird's Eye View lets Dr. Aris glance without reaching for his tablet between rooms. |
| **Smartwatch Alert Integration** | **Dr. Aris** | A wrist notification for 'Patient Ready in Room 3' is the most frictionless possible handoff signal — ideal, but requires additional device pairing and support. |
| **'Next 3 Patients' Queue Preview** | **Dr. Aris** | Showing the next three lobby patients with their chief complaints lets Dr. Aris mentally prepare for upcoming complexity and manage his own cognitive load. |
| **Daily Flow History / Report Export** | **Marcus/Admin** | A post-rush summary of patient throughput and wait times would support Linda's administrative reporting needs, replacing the manual log she currently keeps. |
| **Warm UI Language Customization** | **Linda** | Replacing system language ('Record 502: Active') with warm language ('Welcome, Arthur') is low-effort personalization that aligns with Linda's patient-first values. |
| **Patient Summary Flag("Short Visit" vs "Complex")** | **Provider** | Provider can see at a glance if the next patient will require more time than usual. |
| **"Stuck Room" Flag or "Hold" room button** | **Clinical/Admin** | Admin and Clinical staff have the ability to "Hold" a room or flag a room as "Stuck" if it's been used for longer than 45 mins or needs significant cleaning. |
| **W — Won't Have (this release)** *Explicitly out of scope — may revisit later* | | |
| **Full EHR Chart Editing** | **Dr. Aris** | Deep clinical documentation is out of scope. This tool is a flow coordinator, not an EHR. Adding chart editing risks scope creep and complexity that burdens all three users. |
| **Patient-Facing Self-Check-In Kiosk** | **Linda** | Linda's relationship with patients is a feature, not a bug. A kiosk bypasses the human connection that defines her role and contradicts the clinic's community-oriented identity. |
| **Billing / Claims Processing** | **Linda** | Full billing workflows belong to dedicated practice management software. Including them would clutter Linda's UI with irrelevant alerts — her primary pain point. |
| **Multi-Clinic / Network Dashboard** | **Dr. Aris** | This is a single rural clinic tool. A network-wide view introduces complexity that undermines the clarity and speed every persona in this model requires. |
| **AI-Driven Diagnostic Suggestions** | **Dr. Aris** | Dr. Aris is a specialist. Surfacing AI diagnoses mid-visit would undermine his authority, clutter the HUD, and introduce liability risk outside this project's scope. |

| | | |
|---|---|---|
| **Ability to Log Supplies/Alert for Low supplies to Admin** | **Admin/Clinical** | Out of scope, possibly add an inventory database in the future with this feature. |

The following three themes emerged consistently across every tier of the MoSCoW analysis — appearing in Must Have, Should Have, Could Have, and Won't Have categories. Their recurrence signals that these are not simply individual features but foundational design pillars that must inform every level of the product.

**The HUD (Heads-Up Display) is present in every category — sometimes multiple times within a single tier.** It appears as the Bird's Eye View for Linda, the 5-Second Snapshot for Dr. Aris, and the Clinic Map for Elena. This means the HUD is not a single screen — it is the central nervous system of the entire patient flow model. Every handoff, every status change, and every routing decision runs through it.

**A simple, accessible UI for Linda is featured in all four categories.** Whether it is the one-click check-in (Must), the drag-to-room interaction (Should), warm language customization (Could), or the deliberate exclusion of a self-check-in kiosk (Won't), Linda's need for a low-friction, human-centered interface shapes decisions at every tier. Designing for Linda means designing for trust.

**Staff-to-staff alerts and handoff signals appear in all categories.** The 'Ready for Doctor' button (Must), the Amber Wait Alert and Needs Cleaning status (Should), haptic feedback and smartwatch alerts (Could), and the rejection of a patient kiosk that would bypass staff communication (Won't) all reflect the same underlying need: every member of the clinic team must always know what the others are doing, in real time, without having to ask.

### M — Must Have

| | |
|---|---|
| **1** | **HUD Design and 5-Second Snapshot**<br>The Bird's Eye View and the provider snapshot are two expressions of the same core screen. Getting this right defines the product. All three personas depend on it; all three handoffs flow through it. |

| | |
|---|---|
| **2** | **One-Click Patient Check-In**<br>Linda's check-in interaction is the first moment of truth in the patient experience. Every second of friction here is a second of broken eye contact with a patient who came in expecting to be known. |

| | |
|---|---|
| **3** | **Real-Time Room Status (Color Codes)**<br>The color system is the shared vocabulary of the clinic. Green, Yellow, Purple, Amber — these states must be designed with precision and applied consistently across every device and every user's view. |

### S — Should Have

| | |
|---|---|
| **1** | **Wait Time Amber Alert**<br>This is Linda's most powerful tool for proactive patient management. The moment a name turns amber, she can shift from reactive to empathetic — telling Arthur the doctor is five minutes away before he has to ask. |

| | |
|---|---|
| **2** | **Drag-to-Room Assignment**<br>By mirroring the physical act of moving a patient's chart from one slot to another, drag-to-room collapses Linda's learning curve. It is the interaction design equivalent of 'Don't make me think.' |

| | |
|---|---|
| 3 | **Lab / Referral Order Button**<br>For Dr. Aris, leaving a room to find Elena breaks his focus and the patient's trust. A single in-room dispatch button is a direct investment in his ability to see 20+ patients a day without burnout. |

### C — Could Have

| | |
|---|---|
| 1 | **Voice-to-Text Vitals Entry**<br>Elena's hands are rarely free. Voice input — especially given her bilingual fluency — would allow her to capture vitals conversationally while staying fully present with the patient. |

| | |
|---|---|
| 2 | **Haptic Feedback on Save**<br>The clinic is loud. A quiet vibration confirming a successful save is a small detail with an outsized effect: it gives Elena the confidence to put the tablet down and keep moving without second-guessing herself. |

| | |
|---|---|
| 3 | **'Next 3 Patients' Queue Preview**<br>Dr. Aris is a specialist. Knowing the complexity of his next three patients lets him mentally shift gears between a routine check-up and a post-surgical follow-up — a cognitive advantage that reduces errors and improves care quality. |

## Start with the HUD.

Since the HUD is essential to the model working for all three areas of staffing, it is the correct place to begin design efforts. It is the only element that Linda, Elena, and Dr. Aris all interact with — in different ways, on different devices, but from the same shared data. Getting the HUD right first means every subsequent design decision has a stable foundation to build on.

| Design Dimension | Scope and Design Intent |
|---|---|
| **Look and Color Codes** | Define the full room-status color vocabulary: Green (Clean/Ready), Yellow (In Progress), Purple (Ready for MD), Amber (Wait Alert), Gray (Needs Cleaning). These colors must be accessible (WCAG AA contrast), consistent across desktop, tablet, and wall-mount, and immediately distinguishable in under two seconds. |
| **EHR Integration** | Map the data connections between the HUD and the clinic's existing EHR system. Identify which fields auto-populate the 5-Second Snapshot (patient name, chief complaint, vitals from Elena's entry) and which fields Linda's check-in triggers automatically (insurance status, co-pay balance). The HUD should surface EHR data without requiring any user to open the EHR directly. |
| **Alerts and Notifications** | Design the alert hierarchy: which events trigger a push notification (patient checked in, room ready, wait threshold exceeded), which trigger an in-screen status change (room color shift), and which trigger a passive indicator (Clinic Health delay bar). Map each alert to the persona who receives it and the device it appears on. |
| **Interaction Design** | Define every interaction on the HUD: single-click check-in, drag-to-room assignment, tap to signal 'Ready for Doctor,' and the Undo toast. Prototype each gesture at actual device size — desktop for Linda, tablet for Elena, wall-mount and watch for Dr. Aris. Validate that all touch targets meet a minimum 44x44pt tap area on mobile. |
| **Spatial Layout** | The HUD must mirror the physical layout of the clinic. If Room 2 is on the left in the hallway, it must appear on the left on every screen. Conduct a space-mapping session with Linda to confirm the digital layout matches her mental model before any wireframes are finalized. |